

TAY KAI HAN SAMUEL

PHD CANDIDATE



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PROFILE

I am a PhD student in the Department of the Built Environment at the National University of Singapore (NUS). My research examines smart grids and Virtual Power Plants (VPPs), focusing on simplifying building automation systems for users. I am passionate about energy systems and making the world a greener place through automation and control systems.

WORK EXPERIENCE

ST Engineering - Unmanned & Integrated Systems

PROJECT MANAGER

Aug 2023 - Jul 2026

- Spearheaded multiple projects and enquiries to design and construct military containerized shelters, conducted Heating, Ventilation & Air Conditioning (HVAC) calculations to ensure cooling in the room.
- Implemented Internet of Things (IoT) environment monitoring system by implementing sensors to monitor parameters such as temperature, humidity and smoke in a military container shelter.
- Led an Artificial Intelligence (AI) Research & Development (R&D) project focusing on condition monitoring and predictive maintenance of generator health through vibration, temperature and acoustic sensors, with performance optimization through reinforce learning.
- Managed a diverse portfolio of more than SGD \$25 million spanning bidding, R&D, design-and-build, and maintenance projects.

Keppel Offshore & Marine

PROJECT ENGINEER/ LEAD MECHANICAL ENGINEER

Jul 2019 - Jul 2023

- Keppel Group Scholarship Recipient and Management Trainee
- Mechanical designer of multiple construction project including Russia's first Ice-class LNG Bunkering Vessel, Wind Turbine Installation Vessel, a Renewable Energy Testbed Floating Facility & a 900mw HVDC Offshore Substation
- Designed HVAC systems, performing detailed thermal and ventilation calculations, reviewing air conditioning and ventilation schematic diagrams, and ensuring compliance with local HVAC codes and standards to optimize energy efficiency and maximize thermal comfort.
- Performed thermal and fluid mechanics design calculations to ensure design functionality of on board systems such as fire fighting and sanitary systems.
- Developed automation system philosophies based on sensor monitoring feedbacks.
- Used Revit BIM (Building Information Modeling) software to design and update construction models.
- Generated commissioning procedures to test the systems in the vessel.

EDUCATION HISTORY

National University of Singapore (NUS)

DOCTOR OF PHILOSOPHY CANDIDATE

Aug 2026 - Present

- Awarded NUS Research Scholarship
- Research work in smart grids and Virtual Power Plants (VPPs) cost optimisation

MASTER OF SCIENCE (PROJECT MANAGEMENT)

Aug 2021 - Apr 2023

- Attained Masters Degree in Project Management in National University of Singapore.
- Pursued elective modules including BPS5229 Data Science for Built Environment, EE5713 Modern Power Systems and Smart Grid & ME5207 Solar Energy Systems

AUTODESK REVIT ARCHITECTURE, STRUCTURE & MEP (COMBO COURSE)

Aug 2022

- Completed AUTODESK REVIT ARCHITECTURE, STRUCTURE & MEP (COMBO COURSE) by INNOCOM TECHNOLOGIES (S) PTE LTD

BSC. (HONS) IN MECHANICAL ENGINEERING WITH MINOR IN ECONOMICS

Aug 2015 - Apr 2019

- First Class Honors (CAP: 4.63)
- Participated in Warman Design and Build Robotics Competition in Australia.
- Successfully coded an Augmented Reality (AR) application in C# for my Final Year Project (FYP). Piloted the project in Meta School to provide virtual work skill training for people with Intellectual and Development Disability. Achieved an "A" grade for the FYP.
- Participated in Faculty of Engineering Innovation & Research Award
- Captain of NUS Boxing Team

AWARDS AND ACHIEVEMENTS

Recipient of NUS Research Scholarship	Aug 2026
Dean's List – Semester 2, AY2018/2019	Jun 2019
Dean's List – Semester 1, AY2018/2019	Dec 2018
Recipient of the Keppel Group Scholarship	Jun 2015
Best Officer of Singapore Armed Forces 23 Singapore Artillery "HQ" Battery	Mar 2015

SKILLS AND INTEREST

Language Proficiencies:	English, Mandarin (Written and Spoken)
Computer Software Skills:	Coding in C#, C++ & Python, Scikit-learn, Microsoft Office
Soft Skills:	BIM, System Design, Public Speaking, Leadership, Positive Attitude, Project Management
Other Relevant Skills:	Data Analytics, Machine Vision, Machine Learning, Robotics